INTERNATIONAL MONETARY FUND AND
INTERNATIONAL DEVELOPMENT ASSOCIATION

CAMBODIA

Joint IMF/World Bank Debt Sustainability Analysis 2007[10]

Prepared by the staffs of the International Monetary Fund and
the International Development Association

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A. Introduction

External debt—of which almost 35 percent is owed to the United States and the Russian Federation—constitutes almost 95 percent of Cambodia's public debt. At end-2006, Cambodia's total external public debt was about US$2¼ billion (31 percent of GDP),[12] while domestic debt amounted to 2 percent of GDP, one third of which was denominated in foreign currency. The status of resolution of outstanding debt obligations with the Russian Federation and the United States has not changed since the last joint DSA, conducted in 2006 (Country Report No. 06/264).

This joint WB/IMF debt sustainability assessment concludes that Cambodia's debt is on a sustainable path and that it faces a moderate risk of debt distress. All baseline debt burden indicators remain well below their indicative thresholds, with only two thresholds—the NPV of external debt-to-GDP and the NPV of external debt-to-revenue ratios—being breached briefly in the standard stress tests. Provided debt rescheduling agreements are reached, the risk of debt distress will decline further, though low revenue collections will continue to pose risks to debt sustainability.

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[10] This DSA has been prepared jointly by the IMF and the World Bank staffs based on the framework considered by the Boards of the Fund and the Bank in April 2005 and revised in December 2006. The analysis has benefited from inputs provided by the Asian Development Bank and was discussed with the Cambodian authorities during the 2007 Article IV Consultation.

[12] Cambodia received debt relief from the Fund under the Multilateral Debt Relief Initiative (MDRI) of SDR57 million (US$82 million) in early 2006, as outlined in Press Release No. 05/291 (12/23/2005).

The assessment updates the previous DSA, presented to the IMF's Executive Board at the 2006 Article IV consultation, which concluded that Cambodia's debt at the time was still at high risk of distress. The change occurs for two main reasons: further higher-than-expected GDP growth enhancing long term projections and continued large-scale concessional financing from both traditional donors and new development partners. For this DSA, staff also applied the standard 70 percent discount to the nominal value of the debt to the Russian Federation for the past path of the debt.

The baseline assumes robust growth underpinned by agriculture, garment exports and tourism. Specifically, the analysis is based on the assumption that real GDP growth will be sustained at 6½ percent and exports will grow at around 10 percent a year. Business environment reforms are expected to help improve competitiveness. Foreign aid is assumed to remain broadly stable in nominal U.S. dollar terms through the medium term. Fiscal consolidation is expected to be underpinned by improvements in tax and customs administration. The primary deficit is expected to decline over the medium term, as spending increases only partly offset the improved revenue position. In the longer term, marginal primary surpluses are expected. New borrowing is projected to be somewhat less concessional than the existing debt.

It is expected that Cambodia will start producing oil from around 2011, but because of the uncertainty still surrounding the size of usable deposits and the timing of production, oil receipts are included only in an alternative scenario. Assumptions on oil production are moderate but would favorably impact growth and government revenues from 2011.[13] Under this scenario, real GDP growth would be somewhat higher and government revenues increased substantially. The fiscal deficit would widen somewhat in the early years of oil production to accommodate larger spending for development and poverty reduction. Nevertheless, the overall impact would be to reinforce the favorable trend in debt sustainability.

B. Context

Cambodia remains in arrears on debts to the two largest creditors—the Russian Federation and the United States. Cambodia is not servicing its debt to either creditor, but has made efforts to conclude agreements with both under the framework of the Paris Club. Since prospects for early resolution look remote, the current DSA assumes no restructuring in its baseline, where arrears continue to be built up to the United States and the Russian Federation throughout the projection period.

The current DSA includes the negotiated debt stocks for the debts to the Russian Federation. In discussing debt rescheduling, Cambodia and the Russian Federation have agreed that (i) after applying the stipulated exchange rate and the 70 percent upfront discount, the total debt would be US$457 million (from US$1.5 billion in the previous

[13] An FDI-financed, offshore operation without large domestic investments in shipping facilities or refineries. Production would begin in 2011. Fiscal revenues from oil would reach around US$1.5 billion or around 5 percent of GDP in 2017.

DSA),[14] and (ii) the interest rate for the pre-cut-off-date debt would be about 0.8 percent, with a concessional repayment schedule. However, the countries have disagreed on the classification of loans valued at US$40 million as either pre-cut-off-date or post-cut-off-date debt, and the interest rate on the post-cut-off-date debt. On the latter issue, while Cambodia has stated its desire to have a flat interest rate profile (at 3 percent), the Russian government has expressed its need to have rates approaching market rates at some point in the payment profile. The latest proposals from the two sides are very similar in NPV terms. Despite this, no agreement is in sight.

The United States and Cambodia have reached agreement on the total amount of principal due. After determining the total obligation to be US$162 million, the Cambodian government has received from its U.S. counterpart a draft bilateral agreement based on the 1995 Paris Club agreement (i.e., flow rescheduling on Naples terms assuming a 40-year maturity, 16-year grace period, and an interest rate of 3 percent). The draft agreement is still under review by the Cambodian government. The arrears that would arise as part of that agreement would need to be rescheduled in the context of the Paris Club.

Cambodia has normalized debt relations with other Paris Club creditors. Bilateral agreements were signed with France, Germany, and Japan, and payments resumed. In recent months, Cambodia has repaid outstanding debt owed to Japan under the Paris Club rescheduling, around US$4 million.

Cambodia recently received speculative-grade sovereign credit ratings of B+ from Standard and Poor's and B2 from Moody's, which could provide an opportunity for commercial borrowing. Although no plans have been announced, it is possible that bonds could be issued to finance infrastructure development, possibly for the oil sector. The authorities are also considering BOT operations with Chinese companies, several contracts for which have already been signed. Staffs of the Bank and Fund have not examined the contracts for the guarantees offered; thus, contingent liabilities from such contracts are not incorporated in this DSA. In this context, staffs have maintained that careful assessment is needed to plan for large infrastructure investments, and transactions should be monitored closely and transparently.

Cambodia ranks as a 'weak performer' in terms of its policies and institutions according to the latest World Bank's Country Policy and Institutional Assessment (CPIA). Indicative debt-burden thresholds for countries in this category are a NPV of external debt-to-exports of 100 percent, to GDP of 30 percent and to revenues of 200 percent, and debt service ratios to exports and to revenues of 15 and 25 percent, respectively.

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[14] The previous DSA (in Country Report No. 06/264) incorporated in its analysis the exchange rate to convert USSR Gosbank Ruble debt to U.S. Dollars, but the discount was applied only after the agreement to rescheduling the future path of the debt. The current DSA applies the 70 percent discount for both the past and the future path of the debt, consistent with the 1997 MoU between Russia and the Paris Club and with agreements reached in the context of bilateral negotiations between Cambodia and the Russian Federation.

C. External Debt Sustainability

Cambodia's external debt burden indicators remain below their indicative thresholds, and are likely to decline further, with two thresholds being breached in the standard bound tests. The ratio to GDP of the NPV of external debt will remain below the 30 percent indicative threshold in the medium term, barely breaching the threshold under only one stress test. With a 30 percent nominal depreciation relative to the baseline in 2008, NPV of debt-to-GDP ratio rises by almost 10 percentage points, but declining below the indicative threshold in the medium term. It should be noted, however, that the devaluation test overestimates the negative impact of depreciation since Cambodia is 90-95 percent dollarized.[15] The NPV of debt-to-export ratio, at 32 percent in 2007, is well below its indicative debt burden threshold of 100 percent under all scenarios. Debt service remains below 1 percent of exports throughout the entire projection period even under the most extreme stress test, and is well below the indicative threshold of 15 percent. The external debt-to-revenue ratio is also below the threshold for all years of the baseline, declining from 175 in 2007 to 137 by 2012, and further declining over the long term. It breaches its threshold only in the bound tests that project a sizable increase in debt or a large depreciation. The analysis of external debt in light of the joint guidelines thus shows that Cambodia is now at a moderate risk of debt distress.

Reaching debt rescheduling agreements with the United States and the Russia would further marginally reduce the risk of debt distress. An additional scenario is added to illustrate the case where Cambodia reaches debt rescheduling agreements with the two creditors and begins servicing debts in 2007.[16] It shows that NPV of external debt-to-GDP ratio would drop by 3 percentage points to 19 percent in 2007, and steadily decline to 7 percent of GDP in 2027. The debt service-to-exports ratio remains below 1 percent over the projection period.

D. Public Debt Sustainability

Given that most of public debt is external debt, the public debt dynamics closely follow the external debt dynamics. Under the baseline scenario, the nominal stock of public debt, which amounted to 33 percent of GDP at end-2006, is projected to decline to 30 percent in 2007. The NPV of public sector debt-to-GDP ratio would fall steadily from 23 percent in 2007 to 8 percent by 2027. The NPV of public debt-to-revenue ratio, which was 176 percent at end-2006, is also projected to decline over the projection period. The debt-service-to-revenue ratio also remains low in most scenarios because of the concessionality of the existing debts.

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[15] Contrary to the DSA, which assumes that nominal GDP in US$ would fall by almost as much as the depreciation, a depreciation in the case of Cambodia would affect nominal GDP in US$ only marginally.

[16] For the U.S. debt, the payment schedule based on the latest bilateral agreement sent by the United States (48 consecutive semi-annual back-loaded installments commencing in 2012 and an interest rate of 3 percent) is assumed. For the Russian debt, the disputed US$40 million loans are classified as post-cut-off-date debt, and the interest rates on the post-cut-off-date debt is assumed to rise gradually from 2.5 to 5.75 percent throughout payment period.

An additional bound test is performed to characterize one-time large-scale commercial borrowing to finance either onshore infrastructure development for the oil sector that does not significantly boost growth or a major bailout of the banking sector (Bound test B6 of Table A2).[17] This shows that the debt NPV ratios increase sharply and remain high until 2012. With the interest rate still below nominal GDP growth, the ratios fall over the long term.

Maintenance of prudent fiscal policies is key to continued debt sustainability. The historical scenario—in which real GDP growth and primary deficits are at historical averages (9.8 and 2.0 percent, respectively)—illustrates the case where the government runs significant primary deficits over the long term. Despite continued high GDP growth, deficit financing at less concessional terms leads to rising debt burden indicators. Debt sustainability would be at greater risk if the government borrowed at commercial terms.

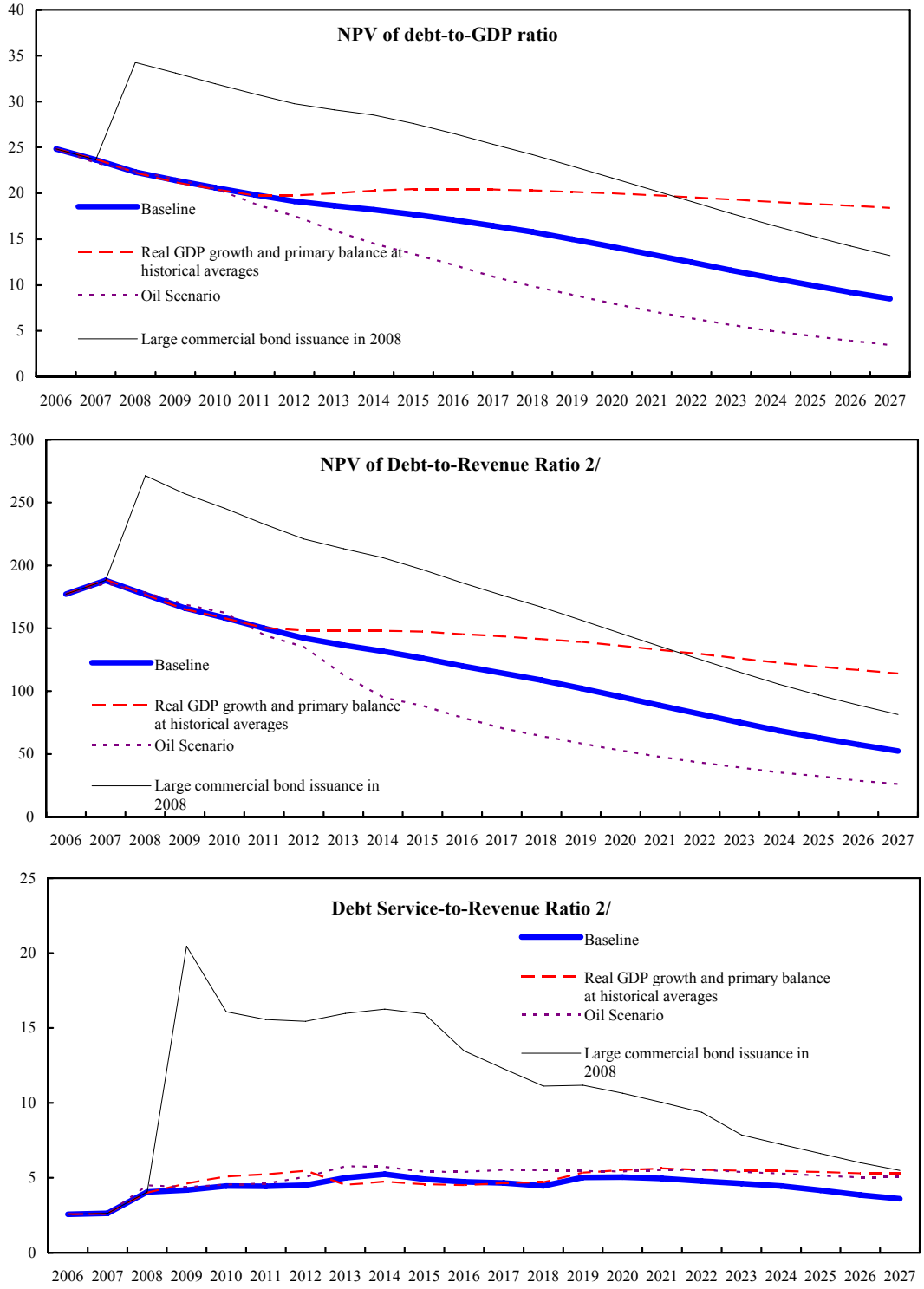
E. Conclusion

Staffs conclude that Cambodia's debt is sustainable, and it faces a moderate risk of debt distress. No distress thresholds are breached in the baseline, with only two thresholds being breached briefly in the standard stress tests. The moderate risk rating is also reinforced by two qualitative considerations: (i) the uncertainty over the contingent liabilities presented by contracts already signed with foreign operators in infrastructure sectors; and (ii) the existence of external arrears to the Russian Federation and the United States. Debt rescheduling agreements, when concluded, will reduce the risk even further if baseline assumptions—improved revenue performance and sustained growth—hold.

The risks to the debt path would seem to be fairly evenly balanced. On the one hand, the positive impact of oil production on growth, exports and fiscal revenues in the longer term would reduce the risks Cambodia faces. This would assume continued prudent economic policies—including strengthening its non-oil revenue base—and export diversification. On the other hand, borrowing against the uncertain future oil revenues could present problems. Staffs also note that while the situation may be sustainable, policy built on arrears accumulation may not be appropriate. Failure to reach the debt agreements over the medium term could also give rise to reputational damage and continue to preclude a financial arrangement with the Fund. Further, close attention needs to be paid to Cambodia's debt management strategy, especially regarding less concessional loans, as it could impair sustainability, especially in the presence of a low revenue base and weak institutions.

[17] This assumes that Cambodia issues a sovereign US$ bond with a face value of 10 percent of GDP in 2008, at 9 percent interest rate and 15-year maturity.

Figure A1. Cambodia: Indicators of Public Debt Under Alternative Scenarios, 2006-2027 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2017 for each of the comparisons.
2/ Revenue including grants.

Table A1. Cambodia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2004-2027
(In percent of GDP, unless otherwise indicated)

	Actual					Projections									
	2004	2005	2006	Historical Average 5/	Standard Deviation 5/	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
Public sector debt 1/	42.9	37.1	33.2			30.7	29.4	28.5	27.5	26.5	25.5		20.1	9.3	
o/w foreign-currency denominated	38.5	33.9	31.0			29.5	28.4	27.7	26.9	26.0	25.0		19.8	9.2	
Change in public sector debt	-0.5	-5.8	-3.9			-2.5	-1.3	-0.9	-1.0	-1.0	-1.0		-1.1	-0.9	
Identified debt-creating flows	-2.0	-5.0	-7.3			-2.4	-1.5	-1.2	-1.2	-1.4	-1.5		-1.2	-0.9	
Primary deficit	2.6	1.0	-0.7	2.0	1.6	1.1	1.6	1.4	1.3	1.1	0.9	1.3	0.3	-0.2	0.2
Revenue and grants	12.0	12.5	14.0			12.6	12.6	12.9	13.0	13.2	13.5		14.4	16.2	
of which : grants	1.7	2.1	2.5			1.8	1.4	1.5	1.3	1.3	1.2		0.8	0.2	
Primary (noninterest) expenditure	14.7	13.5	13.3			13.7	14.2	14.3	14.3	14.4	14.4		14.7	16.0	
Automatic debt dynamics	-5.1	-6.3	-5.4			-3.8	-3.1	-2.8	-2.7	-2.6	-2.5		-1.6	-0.7	
Contribution from interest rate/growth differential	-4.9	-6.1	-4.4			-3.1	-2.6	-2.4	-2.3	-2.2	-2.1		-1.4	-0.6	
of which : contribution from average real interest rate	-0.9	-1.0	-0.8			-0.4	-0.4	-0.3	-0.3	-0.3	-0.3		-0.1	0.0	
of which : contribution from real GDP growth	-4.0	-5.1	-3.6			-2.8	-2.2	-2.1	-2.0	-1.9	-1.9		-1.3	-0.6	
Contribution from real exchange rate depreciation	-0.2	-0.3	-1.0			-0.6	-0.5	-0.4	-0.4	-0.4	-0.4		...	...	
Other identified debt-creating flows	0.4	0.3	-1.1			0.2	0.0	0.2	0.2	0.1	0.1		0.1	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.4	0.3	-1.1			0.2	0.0	0.2	0.2	0.1	0.1		0.1	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.6	-0.8	3.4			-0.1	0.2	0.3	0.2	0.3	0.4		0.1	0.1	
NPV of public sector debt	30.8	27.0	24.8			23.6	22.3	21.4	20.6	19.8	19.1		16.4	8.5	
o/w foreign-currency denominated	26.4	23.8	22.6			22.4	21.3	20.6	20.0	19.3	18.6		16.1	8.4	
o/w external	26.4	23.8	22.6			22.4	21.3	20.6	20.0	19.3	18.6		16.1	8.4	
NPV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	3.2	1.5	-0.3			1.5	2.1	2.0	1.9	1.7	1.5		1.0	0.3	
NPV of public sector debt-to-revenue ratio (in percent) 3/	255.6	216.7	177.2			188.0	176.8	166.1	158.4	150.0	142.1		114.4	52.4	
o/w external	219.1	191.0	161.5			178.1	169.1	160.1	153.7	145.8	138.5		112.3	51.6	
Debt service-to-revenue ratio (in percent) 3/ 4/	4.4	3.7	2.6			2.6	4.1	4.2	4.4	4.4	4.5		4.6	3.6	
Primary deficit that stabilizes the debt-to-GDP ratio	3.1	6.8	3.2			3.8	2.9	2.3	2.3	2.2	2.0		1.5	0.9	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	10.0	13.5	10.8	9.8	2.4	9.1	7.9	7.5	7.5	7.5	7.5	7.9	6.5	6.5	6.5
Average nominal interest rate on forex debt (in percent)	0.7	0.7	0.6	0.5	0.1	0.7	0.7	0.8	0.8	0.9	0.9	0.8	1.4	2.0	1.6
Average real interest rate on domestic currency debt (in percent)	...	...	-5.2	...	...	-4.3	-4.6	-4.6	-4.8	-5.1	-5.2	-4.8	-4.4	-4.4	-4.4
Real exchange rate depreciation (in percent, + indicates depreciation)	-0.6	-0.9	-3.3	1.0	4.3	-2.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	4.8	6.1	4.7	2.4	2.9	4.0	3.9	3.5	3.5	3.6	3.7	3.7	3.3	3.3	3.3
Growth of real primary spending (deflated by GDP deflator, in percent)	-0.5	4.0	9.5	10.0	6.9	12.3	11.9	8.4	7.5	8.0	7.7	9.3	6.7	7.5	7.2
Grant element of new external borrowing (in percent)	...	...	...			43.2	40.8	43.1	37.9	36.3	35.2	39.4	30.3	29.9	...

Sources: Country authorities; and Fund staff estimates and projections.

1/ General government gross debt is used.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

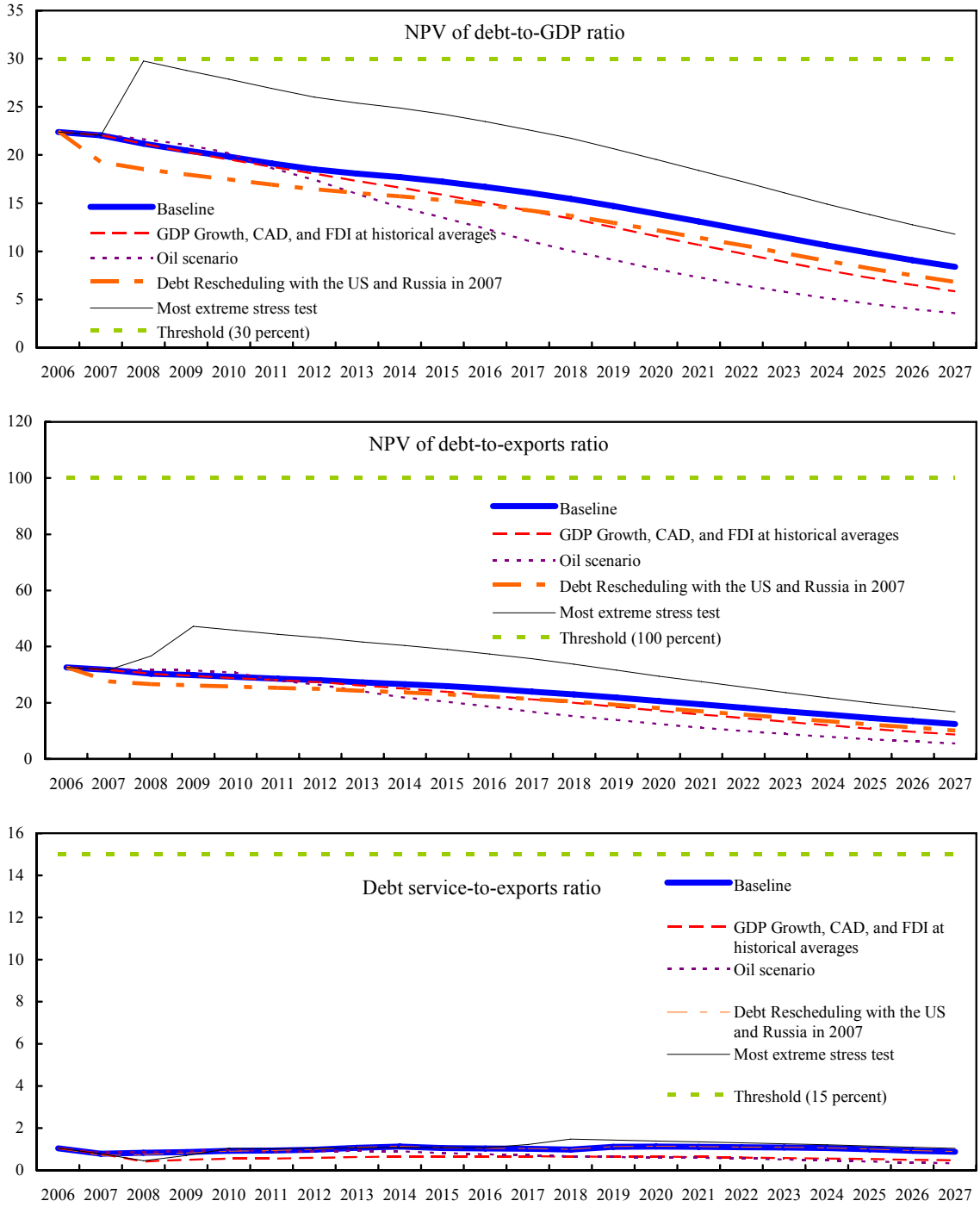
5/ Historical averages and standard deviations are generally derived over the past 8 years, since the consistent balance of payments data are available only since 1999.

Table A2. Cambodia: Sensitivity Analysis for Key Indicators of Public Debt 2007-2027

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of Debt-to-GDP Ratio								
Baseline	24	22	21	21	20	19	16	8
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	24	22	21	20	20	20	20	18
A2. Primary balance is unchanged from 2006	24	22	21	21	20	19	16	8
A3. Permanently lower GDP growth 1/	24	22	22	21	21	20	19	17
A4. Oil Scenario	23	22	22	21	19	18	11	3
B. Bound tests								
B1. Real GDP growth is at baseline minus one standard deviations in 2008-2009	24	23	23	22	22	21	20	14
B2. Primary balance is at baseline minus one standard deviations in 2008-2009	24	24	24	23	22	21	18	9
B3. Combination of B1-B2 using one half standard deviation shocks	24	23	23	22	21	20	17	9
B4. One-time 30 percent real depreciation in 2008	24	31	29	27	25	24	20	11
B5. 10 percent of GDP increase in other debt-creating flows in 2008	24	34	32	31	30	29	22	10
B6. Large commercial bond issuance in 2008	24	34	33	32	31	30	25	13
NPV of Debt-to-Revenue Ratio 2/								
Baseline	188	177	166	158	150	142	114	52
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	189	177	165	158	150	148	144	114
A2. Primary balance is unchanged from 2006	188	177	166	158	150	142	114	52
A3. Permanently lower GDP growth 1/	188	178	168	162	156	150	134	107
A4. Oil Scenario	186	179	169	162	145	135	71	26
B. Bound tests								
B1. Real GDP growth is at baseline minus one standard deviations in 2008-2009	188	182	177	171	165	159	140	87
B2. Primary balance is at baseline minus one standard deviations in 2008-2009	188	187	184	175	165	156	124	57
B3. Combination of B1-B2 using one half standard deviation shocks	188	183	178	169	160	151	121	55
B4. One-time 30 percent real depreciation in 2008	188	243	222	207	192	179	140	67
B5. 10 percent of GDP increase in other debt-creating flows in 2008	188	266	251	240	227	216	152	62
B6. Large commercial bond issuance in 2008	188	271	257	246	233	221	176	81
Debt Service-to-Revenue Ratio 2/								
Baseline	2.6	4.1	4.2	4.4	4.4	4.5	4.6	3.6
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	2.6	4.0	4.6	5.1	5.2	5.5	4.6	5.3
A2. Primary balance is unchanged from 2006	2.6	4.1	4.2	4.4	4.4	4.5	4.6	3.6
A3. Permanently lower GDP growth 1/	2.6	4.1	4.3	4.6	4.7	4.9	5.7	6.7
A4. Oil Scenario	2.6	4.5	4.4	4.5	4.6	5.1	5.5	5.0
B. Bound tests								
B1. Real GDP growth is at baseline minus one standard deviations in 2008-2009	2.6	4.1	4.6	5.2	5.2	5.4	5.8	5.7
B2. Primary balance is at baseline minus one standard deviations in 2008-2009	2.6	4.1	5.8	6.4	5.1	5.0	4.9	4.0
B3. Combination of B1-B2 using one half standard deviation shocks	2.6	4.1	5.1	5.5	4.8	4.8	4.8	3.8
B4. One-time 30 percent real depreciation in 2008	2.6	4.4	4.9	5.2	5.2	5.3	5.6	4.4
B5. 10 percent of GDP increase in other debt-creating flows in 2008	2.6	4.1	13.6	6.7	5.8	5.7	5.4	4.9
B6. Large commercial bond issuance in 2008	2.6	4.1	20.5	16.1	15.6	15.4	12.3	5.5

Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Figure A2. Cambodia: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2006-2027 1/
(In percent)



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2017 for each of the comparisons.

Table A3. Cambodia: External Debt Sustainability Framework, Baseline Scenario, 2004-2027 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2004	2005	2006			2007	2008	2009	2010	2011	2012	2005-10 Average	2017	2027	2012-27 Average
External debt (nominal) 1/	38.5	33.9	31.0			29.5	28.4	27.7	26.9	26.0	25.0		19.8	9.2	
o/w public and publicly guaranteed (PPG)	38.5	33.9	31.0			29.5	28.4	27.7	26.9	26.0	25.0		19.8	9.2	
Change in external debt	-0.4	-4.6	-2.9			-1.5	-1.0	-0.7	-0.8	-1.0	-1.0		-1.1	-0.9	
Identified net debt-creating flows	-4.9	-7.7	-9.1			-7.1	-4.9	-4.0	-3.9	-3.6	-3.3		-4.5	-5.1	
Non-interest current account deficit	1.8	3.9	1.7	2.6	1.3	1.8	3.5	4.1	4.1	4.2	4.3		2.3	0.8	1.8
Deficit in balance of goods and services	7.4	8.7	7.6			6.8	7.8	8.1	7.5	7.0	6.6		2.4	-1.3	
Exports	63.9	64.2	68.8			70.2	70.5	69.5	68.8	67.6	66.4		67.2	67.3	
Imports	71.3	73.0	76.4			77.0	78.2	77.5	76.2	74.6	73.0		69.6	66.0	
Net current transfers (negative = inflow)	-9.4	-8.5	-9.6	-10.4	1.2	-9.1	-8.5	-8.0	-7.4	-6.7	-6.1		-3.9	-1.4	-3.2
Other current account flows (negative = net inflow)	3.8	3.7	3.6			4.1	4.3	4.0	4.0	4.0	3.8		3.9	3.5	
Net FDI (negative = inflow)	-2.3	-6.0	-6.5	-4.2	1.9	-6.7	-6.6	-6.5	-6.4	-6.2	-6.1		-5.9	-5.5	-5.8
Endogenous debt dynamics 2/	-4.4	-5.6	-4.2			-2.2	-1.8	-1.7	-1.6	-1.6	-1.5		-1.0	-0.4	
Contribution from nominal interest rate	0.4	0.3	0.3			0.3	0.3	0.2	0.2	0.2	0.3		0.3	0.2	
Contribution from real GDP growth	-3.4	-4.4	-3.2			-2.4	-2.1	-1.9	-1.9	-1.8	-1.8		-1.2	-0.6	
Contribution from price and exchange rate changes	-1.4	-1.5	-1.4			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	4.5	3.0	6.3			5.6	3.9	3.3	3.1	2.6	2.3		3.4	4.3	
o/w exceptional financing 8/	0.4	0.3	-1.1			0.2	0.0	-0.2	-0.2	-0.1	-0.1		-0.1	0.0	
NPV of external debt 4/	26.4	23.8	22.6			22.4	21.3	20.6	20.0	19.3	18.6		16.1	8.4	
In percent of exports	41.3	37.1	32.9			31.9	30.3	29.7	29.1	28.6	28.1		24.0	12.4	
NPV of PPG external debt	26.4	23.8	22.6			22.4	21.3	20.6	20.0	19.3	18.6		16.1	8.4	
In percent of exports	41.3	37.1	32.9			31.9	30.3	29.7	29.1	28.6	28.1		24.0	12.4	
Debt service-to-exports ratio (in percent) 7/	1.4	1.3	1.1			0.8	0.8	0.9	0.9	0.9	1.0		1.0	0.9	
PPG debt service-to-exports ratio (in percent) 7/	1.4	1.3	1.1			0.8	0.8	0.9	0.9	0.9	1.0		1.0	0.9	
Total gross financing need (billions of U.S. dollars)	0.0	-0.1	-0.3			-0.4	-0.2	-0.2	-0.2	-0.2	-0.2		-0.7	-2.4	
Non-interest current account deficit that stabilizes debt ratio	2.2	8.5	4.5			3.3	4.5	4.8	4.9	5.2	5.3		3.4	1.6	
Key macroeconomic assumptions															
Real GDP growth (in percent)	10.0	13.5	10.8	9.8	2.3	9.1	7.9	7.5	7.5	7.5	7.5	7.9	6.5	6.5	6.5
GDP deflator in US dollar terms (change in percent)	3.7	4.1	4.4	1.3	2.8	5.6	3.9	3.5	3.5	3.6	3.7	4.0	3.0	3.0	3.0
Effective interest rate (percent) 5/	1.1	1.0	1.2	1.1	0.1	0.9	1.0	1.0	1.0	1.0	1.1	1.0	1.5	2.1	1.7
Growth of exports of G&S (US dollar terms, in percent)	28.7	18.7	23.8	23.0	11.3	17.7	12.4	9.8	10.1	9.5	9.6	11.5	10.0	9.7	9.8
Growth of imports of G&S (US dollar terms, in percent)	22.0	20.9	21.1	19.2	8.5	16.2	13.8	10.3	9.4	9.1	9.1	11.3	8.7	9.3	9.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	43.3	40.8	43.2	38.0	36.4	35.3	39.5	30.3	29.9	30.6
Memorandum item:															
Nominal GDP (billions of US dollars)	5.3	6.3	7.3			8.4	9.4	10.4	11.6	12.9	14.4		22.9	57.7	

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 8 years, since the consistent balance of payments data are only available since 1999.
7/ Debt services are cash basis until 2006.
8/ Includes effect of MDRI debt relief.

Table A4. Cambodia: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007-27
(In percent)

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of debt-to-GDP ratio								
Baseline	22	21	21	20	19	19	16	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	22	21	20	20	19	18	14	6
A2. New public sector loans on less favorable terms in 2008-27 2/	22	22	22	21	21	21	19	12
A3. Oil scenario	22	22	21	20	19	17	11	4
A4. Debt Rescheduling with the US and Russia in 2007	20	19	18	18	17	17	14	7
B. Bound Tests								
B1. Real GDP growth at baseline minus one standard deviation in 2008-09	22	22	22	21	20	20	17	9
B2. Export value growth at baseline minus one standard deviation in 2008-09 3/	22	24	29	28	26	25	21	10
B3. US dollar GDP deflator at baseline minus one standard deviation in 2008-09	22	22	21	21	20	19	17	9
B4. Net non-debt creating flows at baseline minus one standard deviation in 2008-09 4/	22	23	24	24	23	22	18	9
B5. Combination of B1-B4 using one-half standard deviation shocks	22	24	27	26	25	24	20	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	22	30	29	28	27	26	23	12
NPV of debt-to-exports ratio								
Baseline	32	30	30	29	29	28	24	12
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	32	30	29	29	28	27	21	9
A2. New public sector loans on less favorable terms in 2008-27 2/	32	31	31	31	31	31	29	17
A3. Oil scenario	32	32	32	31	28	26	17	5
A4. Debt Rescheduling with the US and Russia in 2007	28	27	26	26	25	25	21	10
B. Bound Tests								
B1. Real GDP growth at baseline minus one standard deviation in 2008-09	32	30	30	29	29	28	24	12
B2. Export value growth at baseline minus one standard deviation in 2008-09 3/	32	37	47	46	44	43	36	17
B3. US dollar GDP deflator at baseline minus one standard deviation in 2008-09	32	30	30	29	29	28	24	12
B4. Net non-debt creating flows at baseline minus one standard deviation in 2008-09 4/	32	33	35	34	33	33	27	13.5
B5. Combination of B1-B4 using one-half standard deviation shocks	32	34	39	38	37	36	30	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	32	30	30	29	29	28	24	12
Debt service ratio								
Baseline	0.8	0.8	0.9	0.9	0.9	1.0	1.0	0.9
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	0.8	0.4	0.5	0.5	0.6	0.6	0.7	0.5
A2. New public sector loans on less favorable terms in 2008-27 2/	0.8	0.4	0.5	0.6	0.7	0.7	1.0	1.0
A3. Oil scenario	0.8	0.8	0.9	1.0	0.9	0.9	0.7	0.3
A4. Debt Rescheduling with the US and Russia in 2007	0.7	0.8	0.8	0.9	0.9	1.0	1.0	0.9
B. Bound Tests								
B1. Real GDP growth at baseline minus one standard deviation in 2008-09	0.8	0.4	0.5	0.6	0.6	0.6	0.7	0.7
B2. Export value growth at baseline minus one standard deviation in 2008-09 3/	0.8	0.4	0.7	1.0	1.0	1.0	1.2	1.0
B3. US dollar GDP deflator at baseline minus one standard deviation in 2008-09	0.8	0.4	0.5	0.6	0.6	0.6	0.7	0.7
B4. Net non-debt creating flows at baseline minus one standard deviation in 2008-09 4/	0.8	0.4	0.6	0.7	0.7	0.7	0.9	0.8
B5. Combination of B1-B4 using one-half standard deviation shocks	0.8	0.4	0.6	0.8	0.8	0.8	1.0	0.9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	0.8	0.4	0.5	0.6	0.6	0.6	0.7	0.7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	33	33	33	33	33	33	33	33

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.